

Control work 8 — volume and surface, and work on the mistakes

Twenty-five marks on the solids block, then the diagnosis and the rewriting.

LESSON 155–156

2 × 40 MIN

MATEMATIKA 6, NAZORAT ISHI 8

S7 15 REVIEW

LESSON CLOCK

3

40'

27'

10'

Instructions	3 min
The paper	40 min
Answers and diagnosis	27 min
What comes next	10 min

LEARNING OBJECTIVES

- Find the volume and the surface area of a cuboid and of a cube.
- Run both formulae backwards to a missing edge.
- Solve a capacity, a filling-time and a displacement problem.
- Name each lost mark by its type and rewrite the whole solution.

TEXTBOOK

National	pp. 400–448
Cambridge	Stage 7, pp. 156–172
Workbook	Control paper 8

01

Explanation

The paper — 25 marks, 40 minutes

Q	TASK	MARKS	FROM
1	The volume and surface area of a $6 \times 5 \times 4$ cm cuboid	3	L139, L143–145
2	The volume and surface area of a cube of edge 5 cm	3	L139
3	A cuboid of volume 168 cm^3 on a 7×6 base: the height	3	L140–142
4	A cube of surface area 384 cm^2 : the edge	3	L140–142
5	A cuboid 8 by 3 cm with $S = 158 \text{ cm}^2$: the third edge	4	L140–142
6	A tank $70 \times 40 \times 50$ cm: its capacity, and the time to fill it at 14 l/min	4	L146–149
7	A stone raises the water from 8 cm to 9.2 cm on a 50×30 cm base: its volume	3	L146–149
8	The card for an open box $12 \times 8 \times 5$ cm	2	L143–145

THE ANSWERS

120 cm^3 and 148 cm^2 ; 125 cm^3 and 150 cm^2 ; 4 cm; 8 cm; 5 cm; 140 litres and 10 min; 1800 cm^3 ; 296 cm^2 .

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
a metre left as 1	$1 \cdot 80 \cdot 50$	$100 \cdot 80 \cdot 50$
a cube edge by dividing by three	$125 \div 3$	$a^3 = 125, a = 5$
the root taken before the six	$\sqrt{384}$	$\sqrt{(384 \div 6)} = 8$
litres divided by an area	$140 \div 2800$	$140\,000 \div 2800$
a volume written in cm^2	120 cm^2	120 cm^3
the lid counted on an open box	392 cm^2	296 cm^2
the new level used instead of the rise	$1500 \cdot 9.2$	$1500 \cdot 1.2$

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

What comes next

IF YOU LOST MARKS ON	REVISE
Q1–Q2	the two formulae, L139 and L143–145
Q3–Q5	the formulae run backwards, L140–142
Q6–Q7	capacity, filling and displacement, L146–149
Q8	choosing which faces to count, L143–145

LOOKING FORWARD

Quarter IV opens with data handling — collecting it, drawing pie charts and reading them — then probability, the three-dimensional shapes and their nets, and the revision of the year.

02 Worked examples

EXAMPLE 1 Model answer, Q5: a cuboid 8 cm by 3 cm with surface area 158 cm².

$$① \quad S = 2(8 \cdot 3 + 3c + 8c) = 158.$$

$$② \quad 48 + 22c = 158, \text{ so } 22c = 110.$$

$$③ \quad c = 5 \text{ cm.}$$

Check: $2(24 + 15 + 40) = 158 \checkmark$

ANSWER

5 cm

EXAMPLE 2 Model answer, Q6: a tank 70 cm by 40 cm by 50 cm, filled at 14 litres a minute.

① $70 \cdot 40 \cdot 50 = 140\,000 \text{ cm}^3$.

② That is 140 litres.

Divide by 1000.

③ $140 \div 14 = 10$ minutes.

ANSWER

140 litres and 10 minutes

EXAMPLE 3 Model answer, Q7: the water rises from 8 cm to 9.2 cm on a 50×30 cm base.

① Base area $50 \cdot 30 = 1500 \text{ cm}^2$.

② The rise is $9.2 - 8 = 1.2 \text{ cm}$.

The rise, not the new level.

③ $1500 \cdot 1.2 = 1800 \text{ cm}^3$.

ANSWER

1800 cm³

03 Interactive model

Return Q7 first and ask who multiplied by 9.2; naming the error as “the level instead of the rise” fixes it for the year.

The paper in eight questions

The volume of a $6 \times 5 \times 4$ cm cuboid:

Its surface area:

A cube of edge 5 cm has volume:

A cuboid of volume 168 cm^3 on a 7×6 base is high:

4 cm

42 cm

24 cm

8 cm

A cube of surface area 384 cm^2 has edge:

64

8

19.6

6

A $70 \times 40 \times 50 \text{ cm}$ tank holds:

14 litres

140 litres

1400 litres

140 000 litres

A rise from 8 to 9.2 cm on 1500 cm^2 means:

13 800 cm^3

1800 cm^3

12 000 cm^3

1.2 cm^3

An open box $12 \times 8 \times 5$ cm needs:

392

296

200

96

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Volume	Hajm	Объём
Surface area	Sirt yuzasi	Площадь поверхности
Capacity	Sig'im	Вместимость
Displacement	Siqib chiqarish	Вытеснение
Open box	Qopqog'i yo'q quti	Открытая коробка
Unit	O'lchov birligi	Единица измерения
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 wants the surface area in:

cm

cm²

cm³

litres

Q4 divides by six:

after the root

before the root

never

twice

Q5 gives an equation that is:

quadratic

linear

impossible

a fraction

Q6 needs the volume converted to:

cm^2

litres

metres

nothing

Q7 multiplies the base area by:

the new level

the rise

the old level

their sum

Q8 leaves out:

the base

the lid

two walls

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The volume of a $6 \times 5 \times 4$ cm cuboid

ANSWER 120 cm^3

2. Its surface area

ANSWER 148 cm^2

3. The volume of a cube of edge 5 cm

ANSWER 125 cm^3

4. Its surface area

ANSWER 150 cm^2

5. $V = 168 \text{ cm}^3$ on a 7×6 base: the height

ANSWER 4 cm

6. One face of a cube of surface area 384 cm^2

ANSWER 64 cm^2

7. The capacity of a $70 \times 40 \times 50$ cm tank

ANSWER 140 litres

Medium · the standard question

7 PROBLEMS

1. The edge of a cube of surface area 384 cm^2

ANSWER 8 cm

2. A cuboid 8 by 3 cm with $S = 158 \text{ cm}^2$: the third edge

ANSWER 5 cm

3. A 140 -litre tank filled at 14 l/min

ANSWER 10 min

4. A stone raising 8 cm to 9.2 cm on a 50×30 base

ANSWER 1800 cm^3

5. Card for an open box $12 \times 8 \times 5 \text{ cm}$

ANSWER 296 cm^2

6. The closed box of the same size

ANSWER 392 cm^2

7. The volume of that box

ANSWER 480 cm^3

Hard · stretch and reason

7 PROBLEMS

1. A 1 m by 80 cm by 50 cm tank, in litres

ANSWER 400 litres

2. The commonest lost mark in this paper

ANSWER A unit not converted

3. A cube of volume 1728 cm^3 : its edge and surface area

ANSWER 12 cm and 864 cm^2

4. The mass of the water in a full 140-litre tank

ANSWER 140 kg

5. Doubling every edge of the Q1 cuboid: the new volume

ANSWER 960 cm^3

6. ...and the new surface area

ANSWER 592 cm^2

7. The depth of 28 litres in a tank of base $70 \times 40 \text{ cm}$

ANSWER 10 cm

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on in full, with the units on every line.

1. Rewrite in full every question on which you lost a mark, naming the slip in the margin.
2. Find the volume and the surface area of a cuboid 7 cm by 6 cm by 3 cm.
3. A cube has volume 1331 cm^3 . Find its edge and its surface area.
4. A tank 100 cm by 60 cm by 45 cm is filled at 15 litres a minute. How long does it take?
5. How much card is needed for an open box 15 cm by 10 cm by 6 cm deep?